Hanyi Seok

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Research Interests: Electrochemical carbon capture and conversion, hydrogen production, and sustainable aviation fuel synthesis

EDUCATION

Dual-degree program between Stony Brook University and the University of Seoul

Stony Brook University *Expected May 2027*
B.E. in Engineering Science, specialization in Environmental Engineering
Second Major in Applied Mathematics and Statistics
GPA: 3.96/4.00

University of Seoul *Expected May 2027*
B.E. in Environmental Engineering
GPA: 4.10/4.50

RESEARCH EXPERIENCE

Seo Research Group **Stony Brook, NY**
Undergraduate Researcher | Advisor: Prof. Hyowon Seo *May 2025 – Present*

Hollow-Fiber Direct Air Capture Modeling

- Developing a hollow-fiber membrane contactor model for OH^-–based direct air capture by reformulating carbonate chemistry in conservative variables (DIC and alkalinity) to predict pH and CO_2 uptake.
- Building a coupled transport model that links fiber-side reactive transport, shell-side CO_2 depletion, and tank mass balance, using flow-weighted outlet concentrations and quasi-steady-state solution methods.
- Calibrating the external mass-transfer coefficient from pH/alkalinity data and preparing a first-author manuscript based on the modeling results.

FeEDDHA Immobilization for Preliminary Electrochemical Studies

- Investigated FeEDDHA immobilization on carbon felt and Ni-based electrodes via soaking, drop-casting, and surface functionalization.
- Assessed catalyst stability and electrochemical behavior using cyclic voltammetry under N_2 and CO_2 in aqueous and DMF media.
- Diagnosed catalyst loss and film delamination as key interfacial stability challenges and screened follow-up conditions for improved catalyst retention.

Water Environment and Engineering Lab **Seoul, South Korea**
Undergraduate Researcher | Advisor: Prof. Ihnsup Han *May 2024 – Jan 2025*

- Investigated treatment of amine-rich wastewater (ARWW) from wet carbon capture using integrated electrochemical oxidation, coagulation, and adsorption.
- Systematically evaluated key process variables, including electrode material, activated carbon dosage, voltage, electrode spacing, influent strength, and temperature, to optimize MEA removal.
- Derived process-design guidelines and identified a cost-effective operating window for ARWW management in the Yongin Semiconductor Industrial Complex.

Environmental Systems Toxicology Lab **Seoul, South Korea**
Undergraduate Researcher | Advisor: Prof. Jinhee Choi *Jun 2021 – Feb 2022*

- Analyzed correlations between in vitro ToxCast assay activity and in vivo ToxRef reproductive-toxicity endpoints to identify candidate screening assays.
- Integrated and cleaned ToxCast and ToxRef datasets covering 449 chemicals and 949 assays; converted assay activity into binary indicators and applied endpoint-specific Pearson correlation analysis.
- Identified two candidate ToxCast assays, NVS_GPCR_rV1 and NVS_GPCR_hM3, as potential predictors of reproductive toxicity.

PRESENTATIONS

H. Seok and D. Jeong. *Process Optimization for Treatment of Amine-Rich Wastewater Generated from Wet Carbon Capture.* Undergraduate Thesis Presentation, Department of Environmental Engineering, University of Seoul, South Korea, 2025.

H. Seok, S. Jang, and Y. Baek. *SELFIES ALPHA: A Stereochemistry-Aware Molecular String Representation for Toxicity Prediction.* Poster, Environmental Health Policy Idea Competition, South Korea, 2023.

H. Seok, M. Jo, and S. Ahn. *Identification of ToxCast Assays for Reproductive Toxicity Screening by Correlation Analysis of ToxCast and ToxRef Data.* Research Workshop, Graduate School of Chemical Safety Management, University of Seoul, South Korea, 2022.

SELECTED PROJECTS

Development of a Dissolved Oxygen Prediction Model for the Han River

Smart Water Environment Management *Sep 2024 – Dec 2024*

- Developed a dissolved oxygen (DO) prediction model for the Han River system using hourly water-quality data and five predictors: temperature, pH, total nitrogen, total phosphorus, and total organic carbon.
- Compared multiple missing-data imputation methods and identified ensemble imputation as the most effective preprocessing strategy for environmental time-series data.
- Benchmarked machine-learning and deep-learning models and optimized a PatchTST model, achieving $R^2 = 0.915$ on test data and 0.966 on unseen November 2024 data; the project was recognized as outstanding among 100 students and presented at the course showcase.

Domestic SAF Roadmap for Korea

Unit Design of Processes in Environmental Engineering *May 2024 – Jul 2024*

- Led an eight-member team to develop a domestic SAF roadmap for Korea, reviewing ASTM-approved production pathways and EU policy requirements for SAF blending.
- Evaluated domestic feedstocks for both bio-based and synthetic SAF pathways, including rice straw, used cooking oil, animal manure, and sewage sludge, and assessed DAC as an additional CO_2 source when required.
- Built a Python model to estimate SAF output and project domestic jet-fuel demand through 2050; proposed an implementation strategy that ranked 1st among 8 teams.

HONORS & AWARDS

Tau Beta Pi Engineering Honor Society	*Fall 2025*
Dean's List, Stony Brook University	*Spring 2025; Fall 2025*
Korea–U.S. Advanced Technology Youth Exchange Program Scholarship (KIAT)	*Spring 2025 – Fall 2025*
Presidential Scholarship, University of Seoul	*Spring 2021 – Fall 2024*
Outstanding Student Scholarship, Digital Water Industry Innovation Talent Development Program	*Spring 2024*
Gold Prize, 2023 Environmental Toxicology and Health Project Competition (1st of 15 teams)	*Nov 2023*
Graduate School Dean's Award, University of Seoul (2nd of 20 teams)	*Feb 2022*

TECHNICAL SKILLS

Programming: Python, R

Modeling & Quantitative Analysis: transport modeling, mass-balance modeling, parameter calibration, statistical analysis

Electrochemistry & Experimental Methods: cyclic voltammetry, electrochemical oxidation, electrode preparation, catalyst immobilization, drop-casting, surface functionalization

Data Analysis & Forecasting: time-series forecasting, missing-data imputation, correlation analysis, machine learning, deep learning

CAD & Prototyping: SolidWorks, Fusion 360, 3D printing

ADDITIONAL EXPERIENCE

Republic of Korea Army **Yangpyeong, South Korea**

Data Analyst | Squad Leader *Apr 2022 – Oct 2023*

- Managed administrative datasets and maintained DELIS, a Korean military administrative management system, to support standardized reporting and personnel operations.
- Automated Python-based data pipelines for cleaning, validating, and aggregating operational records, improving the consistency of weekly reporting workflows.
- Built anomaly-detection and forecasting tools for personnel and operations data and led a 12-soldier squad responsible for training schedules, safety compliance, and performance evaluations.